



DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
West Point, New York 10996

MADN-CBE

22 March 2026

MEMORANDUM FOR RECORD

SUBJECT: Chemical Engineering Program Course and Program Brief (AY27-1)

1. The senior faculty of the chemical engineering program at the United States Military Academy presented a course and program brief to leadership and faculty of the Department of Chemical & Biological Science & Engineering (CBSE) on 20 March 2026 to conduct a pre-semester review of courses to be taught in the Fall of 2026 with a focus on course assessment from AY26-1. The chemical engineering course directors for courses in AY27-1 include Prof. Andrew Biaglow, Dr. Simuck Yuk, Dr. Enoch Nagelli, and LTC Stephen Hummel. Junior chemical engineering faculty in attendance include MAJ Nijel Rogers, CPT(P) Elizabeth Golonski, CPT(P) Chris Stewart, and CPT Denis Gilinski. Chemical engineering course directors presented a summary of the topics of the course, graded events assessment, and the overall improves/sustains together with impacts on lab program support each respective fall semester course to the CBSE leadership, biology program leadership and faculty, and chemistry program leadership and faculty. Chemical engineering faculty members not in attendance were MAJ Louis Tobergte, MAJ Samuel Lowell, and MAJ Josh Frey.

2. Each course director presented topics and relevant assessments to their specific courses. These topics are outlined in more detail in the individual course assessment sections of the course slide. The topics discussed serve as a tool to gauge effectiveness of course content and administration in meeting the ABET student outcomes specific to each course. Comments and questions from CBSE faculty support the department and help provide insight into the chemical engineering major courses, academic coherence, and provide CBSE leadership a more detailed insight into alignment of department goals. Focus areas for course improvement and potential future changes from course assessments are the focus of the general discussions.

3. The following courses were discussed: CH350 Bioprocess Engineering, CH363 Separations Processes, CH365 Chemical Engineering Thermodynamics, CH459 Chemical Engineering Laboratory, and CH485 Heat and Mass Transfer were reviewed and thoroughly discussed by all attendees. Some of the topics led to broader discussions relevant to the entire program. These included: 1) Improvement of math fundamentals for our chemical engineers, 2) Importance of laboratory experiments to reinforce learning, 3) Effort and development of the experiment labs for the CH363 course, and 4) Faculty recruiting and 6-year teaching slate.

4. The point of contact for this document is the undersigned at



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AY27-1 Chemical Engineering Program Brief



19 March 2026

Agenda

- Course Briefings
 - CH350
 - CH363
 - CH365
 - CH459
 - CH485
- 6-Year Teaching Slate
- Future Faculty

CH350: Bioprocess Engineering

Course Director: LTC Hummel

Course Supervisor: Dr. Yuk

Credit Hours: 3.0 (BS=1.0, ET=2.0, MA=0)

Prerequisites: CH102, MA205, and PH202

Co-requisite: None

Lessons: 30 @ 75 min

Special Requirements: None

The purpose of this course is to provide the introductory knowledge for understanding the biotechnology/bioprocessing engineering. Topic includes enzyme kinetics, molecular biology, cell growth, bioreactors, and bioprocesses. The bioprocess control and its application to different biological systems are covered in the classroom instruction. An important emphasis is made on the use of kinetics and process controls on the biological systems for engineering application.

Topics – by Chapter

- 1) *Bioprocess Engineering Basic Concepts*, 3rd Ed., by Michael L. Shuler, Fikret Kargi, Matthew DeLisa, Prentice Hall.
- 2) *Quantitative Fundamentals of Molecular and Cellular Bioengineering*, by K. Dane Wittrup, Brice Tidor, Benjamin J. Hackel, and Casim A. Sarkar, The MIT Press.

- Introduction / Gen. Bio (Ch 1, Shuler et al)
- Enzyme Kinetics (Ch 2-3, Shuler et al; Ch 2-3, Wittrup et al)
- Central Dogma to Molecular Biology (Ch 4-5, 8, Shuler et al)
- Cell Growth (Ch 6-7, Shuler et al)
- Bioreactor Selection (Ch 9-10, Shuler et al)
- Bioprocess Consideration (Ch 11-13, Shuler et al)

Course Assessment – Items from Section III

Sustain:

1. WPR / TEE – required to access the Cadets' understanding of overall course contents. (*TEE Average: 81%*)
2. Capstone Project – development of bioreactors based on the material balance applied for cellular system. (*Project Average: 88%*)

Improve:

1. General – continue to adjust Problem Sets and WPRs
2. Integrate more material from other courses: CH375, CH473, CH485 and CH363 (*WPR 1 average was 71% but increased to 84% in WPR 2 as material shifted → incorporate more biochem / biology throughout the course*)

Assessment – Graded Events

Requirement	# of Event	Points/Event	Total Points	%
Problem Set	6	50	300	20
Instructor Points	4-5	Varies	50	3
Capstone IPR	1	50	50	3
Capstone Presentation	1	100	100	6
Capstone Paper	1	200	200	14
WPR	2	200	400	27
TEE	1	400	400	27
TOTAL (*Individual Points = 1500 (100%))			1500	100

CH363: Separation Processes

Course Director: Dr. Nagelli

Course Supervisor: Dr. Biaglow

Credit Hours: 3.5 (BS=0, ET=3.5, MA=0)

Prerequisites: CH362

Co-requisite: None

Lessons: 30 @ 75 min, 7 @ 120 min

Special Requirements: None

This course covers methods for the physical separation of chemicals. Topics include dew point and bubble point calculations, adiabatic flash, distillation, chromatography, liquid-liquid and gas-liquid absorption/stripping. Students are taught the significance of staging of unit operations. Heavy emphasis is placed on theory of operations, numerical methods of solution, and simulation.

Topics – by Chapter

Separation Process Principles, 4th Ed., by J.D. Seader, E.J. Henley and D.K. Roper

- Introduction to Separations/DOF Analysis (Ch. 1)
- Vapor-liquid, gas-liquid, solid-liquid, flash(Ch. 4)
- Cascading configurations (Ch. 5)
- Designing trayed towers and packed columns (Ch. 6)
- Optimizing towers and columns (Ch. 7)
- Liquid-liquid extraction (Ch. 8)
- Multi-component distillation (Ch. 9)
- Capstone Project

Course Assessment – Items from Section III

Sustain:

- Coverage - Maintain strategy of incorporating McCabe Thiele/NTU/CHEMCAD in as many lessons/topics as possible.
- Blocks & Lesson Structure

Improve:

- **Course Content** - Introduce more instructor problems for problem sets and in class problems.
- **Resources & Laboratories** - Perhaps more tours of CH459 lab and using actual data from the lab for absorber and distillation problems along with NTU calculations. *The average for the WPRs continues to be >85% similar to AY25-1 and the TEE average was 87%. Shows room for improvement.*

Lab Requirements – Will integrate Labs (2x) to be SB Unit Operations.

Assessment – Graded Events

8 Problem Sets @ 30 pts each:	240	14%
3 *In-Class Prob. Sets @ 100 pts each:	300	14%
3 *WPRs @ 200 pts each:	600	28%
7 *Labs @ 30 pts each:	210	10%
1 *Term End Exam @ 500 pts:	500	23%
1 Capstone	300	14%
Total:	2150	
*Individual Points :	1610	75%

CH365: Chemical Engineering Thermodynamics

Course Director: Dr. Biaglow

Course Supervisor: Dr. Yuk

Credit Hours: 3.0 (BS=0, ET=3.0, MA=0)

Prerequisites: CH363, CH364, MA366, MC312

Co-requisite: None

Lessons: 40 @ 55 min

Special Requirements: None

This course covers the body of thermodynamic knowledge necessary for understanding modern chemical process simulation. Students learn the theory behind the thermodynamic methods used in the software. The course includes calculus- and numerical-based thermodynamics approaches for determining the properties of substances, solutions, and multiphase mixtures. Topics include equations of state, pure component properties, transport properties, properties of mixtures, fugacity, excess properties, activity coefficients, and phase equilibria. The problems in the course emphasize engineering applications. Topics covered in class are related to real systems through the use of chemical process simulators.

Topics – by Chapter

Chemical Engineering Thermodynamics, Smith, van Ness, Abbott, and Swihart, 9th Edition (2021)

- Introduction (Ch. 1)
- First Law (Ch. 2)
- Equations of State (Ch. 3)
- Heat (Ch. 4)
- Entropy and Second Law (Ch. 5)
- Fluid Properties (Ch. 6)
- Equilibrium (Ch. 10)
- Solution Thermodynamics (Ch. 11)

Course Assessment – Items from Section III

Sustain:

Writing project – intense grading, but introspective for cadets.

Capstone project – Calculate solution props. & comp. to CC.

Student self-correction in grading.

Improve:

Continue to develop creative ways to emphasize Ch 11, specifically excess properties. Calculate activity and excess properties but do *interpret* them. Activity/Excess Gibbs energy scores on TEE A (77.4% AY26 vs. 75.6% in AY25).

Graded Requirements – Grading rubric for capstone project.

Lab Requirements – None.

Assessment – Graded Events

1 *Term End Exam @ 500 pts:	500	21.01%
1 *Capstone Design Project @ 300 pts	300	12.61%
1 *Capstone IPR, pass/fail:	0	0.00%
3 *WPRs @ 200 pts each:	600	25.21%
66 *Problems @ 10 pts each:	660	27.73%
1 *SIS Quiz @ 60 pts:	60	2.52%
1 *Mathematica Quiz @ 60 pts:	60	2.52%
1 *Writing Assignment @ 200 pts:	200	8.40%
Total:	2380	100.00%
*Individual Points:	2380	100.00%

CH459: Chemical Engineering Lab

Course Director: Dr. Yuk

Course Supervisor: Dr. Nagelli

Credit Hours: 3.5 (BS=0, ET=3.5, MA=0)

Prerequisites: CH362, CH363, and CH364

Co-requisite: None

Lessons: 40 @ 120 min, 7 @ 120 min

Special Requirements: None

This course provides laboratory experience in selected chemical engineering unit operations, such as gas absorption, evaporation, distillation, liquid-liquid extraction, cooling tower, heat exchanger, and chemical reactors. Process control and process safety are emphasized in laboratory and classroom instruction. Written and oral reports required.

Topics – by Chapter

1. *Unit Operations of Chemical Engineering*, 7th Ed., by Warren L. McCabe, Julian C. Smith and Peter Harriott; McGraw-Hill, 2005.
2. *Plant Design and Economics for Chemical Engineers* 5th Ed., Peters, Max S. and Klaus D. Timmerhaus, McGraw-Hill, New York 2003, ISBN-10: 0071240446.

- Batch and CSTR
- Cooling Tower
- Hydrogen Fuel Cell
- Single-Effect Evaporator
- Distillation Column
- Carbon Dioxide Absorber

Course Assessment – Items from Section III

Sustain:

1. Integration of SWE – DIST EXSUM (Individual Submission).
2. Update guidance documents with specific direction for calculations.
3. Continue to update HWs and WPRs for each experiment – Pull from textbooks and new problems.

Improve:

1. Focus Laplace Block Diagram where controls is obvious for cadets conducting experiments (*especially for Round Robin 2; Laplace block transform scores on TEE (84.0% AY26 vs. 90.0% in AY25).*)
2. Perform the walk-through of lab equipment for each round robin, so cadets can have better understandings of equipment design and workflow.

Lab Requirements – 2 Sections C1D1 & E1F1 (Mr. Mathew Support)

Assessment – Graded Events

6 Pre-Lab HWs @ 25 pts ea.	150*	4.0%
1 Exec. Sum + Lab (SWE)	200*	6.7%
1 SWE IPR	25*	0.83%
5 Exec. Sum/Poster/Report @ 100 pts	500	16.7%
5 IPRs @25 pts ea.	125	4.16%
5 Lab Execution @ 100 pts ea.	500	16.7%
2 WPRs @ 400 pts each:	800*	26.7%
1 Term End Exam @ 700 pts:	700*	23.3%
Total:	3000	
*Individual Points :	1875	62.5%

CH485: Heat and Mass Transfer

Course Director: LTC Plante

Course Supervisor: Dr. Biaglow

Credit Hours: 3.5 (BS=0, ET=3.5, MA=0)

Prerequisites: MA364 and ME362

Co-requisite: None

Lessons: 30 @ 75 min, 7 @ 120 min

Special Requirements: None

This course includes the study of the mechanisms of energy and mass transport, with special emphasis on applications in engineering systems. Coverage includes Fourier's Law of Heat Conduction, and Fick's Law of Diffusion, the development of shell energy and species balances, and the use of these equations to solve for temperature and concentration profiles in chemical engineering systems. An important emphasis in the course is the use of transport equations to understand species diffusion, convection, and chemical reaction in equipment design.

Topics

Mass and Heat Transfer, T.W. Fraser Russell, A.S. Robinson, and N.J. Wagner, Cambridge University Press, 2008

- Introduction (1 Lesson)
- Macroscopic mass, energy, and species balances
 - Chemical reactors (2 lessons)
 - Heat exchangers (4 lessons)
 - Mass contactors (2 lessons)
- Microscopic mass, energy, and species balances
 - Conduction, diffusion, and radiation (9 lessons)
 - Special cases (curved geometries, composites, membrane diffusion, transient heat & mass transfer) (4 lessons)
 - Convective heat and mass transfer (7 lessons)
 - Transport Analogies (4 lessons)

Course Assessment – Items from Section III

Sustain:

- Problem solving days prior to WPR/ICPS
- Lesson structure & content (radiation heat transfer)
- Continue Arnold Cell Lab (two different cells)
- Convective H/T Lab – continue concepts; get hands-on

Improve:

- Introduce cooling fin experiment (hands-on) for Lab 3 (*Lab3 score in AY26-1 was 94%, however, the score for the convective transport lab was the lowest of all labs at 92%*)
- Build bench of alternate labs
 - Wetted wall column (*PS3 on mass transfer had lowest average 87%*)
 - Membrane gas diffusion (air separation) (*WPR2C on Gas Diffusion was lowest with average of 76%*).

Lab Requirements – 2 Sections S2 & T2 Hours (Mr. Mathew Support)

Assessment – Graded Events

10 Problem Sets @ 50 pts each:	500	22%
2 In-Class Prob. Sets @ 100 pts each:	200	9%
3 WPRs @ 200 pts each:	600	26%
1 Writing Assignment @ 200 pts:	225	10%
7 Labs @ 40 pts each:	280	12%
1 Term End Exam @ 500 pts:	500	22%
Total:	2305	
Individual Points:	2025	88%

6-Year Teaching Slate

• **ChemE Majors:**

- 25 (Class '26)
- 28 (Class '27)
- 26 (Class '28)
- 31 (Class '29)

ABET visit in Fall AY27-1

- 30AUG-01SEP26

• **Next PhDs:**

LTC Plante (AY27)

- *Cornell U - BioE*

LTC Corrigan (AY28)

- *U of Wash - BioE*

CPT Hudson (AY29)

- *Ga Tech - ChemE*

MAJ Bowers (AY30)

- *Purdue - ChemE*

*** 1st Year for Faculty**

*** MAJ Tobergte possible IMA**

AY27		AY28		AY29	
AY27-1 (Fall)	AY27-2 (Spring)	AY28-1 (Fall)	AY28-2 (Spring)	AY29-1 (Fall)	AY29-2 (Spring)
CH363 (Nagelli)	CH362 (Rogers)	CH363 (Nagelli)	CH362 (Golonski)	CH363 (Nagelli)	CH362 (Jr. Rot.)
CH459 (Yuk)	CH364 (Plante)	CH459 (Corrigan)	CH364 (Plante)	CH459 (Yuk)	CH364 (Plante)
CH485 (Plante)	CH402 (Biaglow)	CH485 (Plante)	CH402 (Biaglow)	CH485 (Plante)	CH402 (Biaglow)
CH365 (Biaglow)	CH400 (Tobergte)	CH365 (Biaglow)	CH400 (Yuk)	CH365 (Biaglow)	CH400 (Yuk)
CH350 (Hummel)	CH367 (Nagelli)	CH350 (Hummel)	CH367 (Nagelli)	CH350 (Corrigan)	CH367 (Nagelli)
	CH300 (Hummel)		CH300 (Glinski)		CH300 (Hudson)
	CH450 (Yuk)	*Yuk Sabbatical	CH450 (Corrigan)		CH450 (Corrigan)
Golonski (GC)	Yuk & Tobergte (1x GC)				
Rogers (GC)	Golonski (GC)	Golonski (GC)			
Frey (GC)	Frey (GC)	Breed(GC)	Breed (GC)	Hudson (GC)	
Stewart (ACD-GC)	Stewart (ACD-GC)	Glinski (GC)	Turner (GC)	Breed (GC)	Breed (GC)
Glinski (GC)	Glinski (GC)	Stewart (CD-GC)	Stewart (CD-GC)	Turner (GC)	Turner (GC)
Breed (GC)	Breed (GC)	Turner (GC)	Glinski (GC)	Brown (GC)	Brown (GC)
AY30		AY31		AY32	
AY30-1 (Fall)	AY30-2 (Spring)	AY31-1 (Fall)	AY31-2 (Spring)	AY32-1 (Fall)	AY32-2 (Spring)
CH363 (Nagelli)	CH362 (Jr. Rot.)	CH363 (Nagelli)	CH362 (Jr. Rot.)	CH363 (Nagelli)	CH362 (Jr. Rot.)
CH459 (Yuk)	CH364 (Bowers)	CH459 (Yuk)	CH364 (Bowers)	CH459 (Yuk)	CH364 (Bowers)
CH485 (Bowers)	CH402 (Biaglow)	CH485 (Bowers)	CH402 (Biaglow)	CH485 (Bowers)	CH402 (Biaglow)
CH365 (Biaglow)	CH400 (Yuk)	CH365 (Biaglow)	CH400 (Yuk)	CH365 (Biaglow)	CH400 (Yuk)
CH350 (Corrigan)	CH367 (Nagelli)	CH350 (Corrigan)	CH367 (Nagelli)	CH350 (Corrigan)	CH367 (Nagelli)
	CH300 (Hudson)		CH300 (Hudson)		CH300 (Corrigan)
	CH450 (Corrigan)		CH450 (Corrigan)		CH450 (Corrigan)
Hudson (ChemE)					
Brown (GC)	Brown (GC)	Hudson (ChemE)			
Turner (GC)	Turner (GC)	Brown (GC)	Brown (GC)		

Future Inbound Faculty

Chem. E. future faculty updates

- LTC Luke Plante (AY27)...sponsor: Dr. Nagelli
- CPT Austin Breed (AY27)...sponsor: CPT Stewart
 - *Northeastern University, M.S. ChemE*
- LTC Trevor Corrigan (AY28)...sponsor: Dr. Yuk
- CPT Madison Turner (AY28)..sponsor: CPT Glinski
 - *Duke University, M.S. BioE*
- CPT Benjamin Hudson (AY29)..sponsor: LTC Plante
- CPT Mitchell Brown (AY29)..sponsor: CPT Turner
- MAJ Patrick Bowers (AY30)...sponsor: LTC Corrigan

Questions?